

Platform Keperawatan Indonesia

Comprehensive Application Flow

Dokumentasi komprehensif seluruh alur (flow) aplikasi Platform Keperawatan Indonesia. Mencakup arsitektur sistem, ERD, 8 peran pengguna, alur rekrutmen, manajemen legalitas STR, sertifikasi kompetensi, career events timeline, API integration, webhook system, dan seluruh business logic yang ada di dalam codebase.

FRAMEWORK	MODELS	CONTROLLERS	ENDPOINTS	TANGGAL
Laravel 11	31 Eloquent	43 Total	140+ Routes	Juli 2026

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1 Arsitektur Sistem & Technology Stack

1.1 Technology Stack

LAYER	TEKNOLOGI	KETERANGAN
Framework	Laravel 11 (PHP)	Full-stack MVC framework
Frontend	Blade + Tailwind CSS + Vite	Server-side rendering dengan modern CSS
Database	SQLite (dev) / PostgreSQL (prod)	Dual-driver support
Queue	Laravel Queue	Sync/database driver
Auth	Laravel Breeze	Session-based authentication
API	REST API v1	API Key + HMAC Signature
Primary Key	UUID (v4)	Semua model menggunakan UUID
Identity Link	NIK (16 digit)	Foreign key utama lintas tabel

1.2 Struktur Direktori Aplikasi

Controllers (43 total)

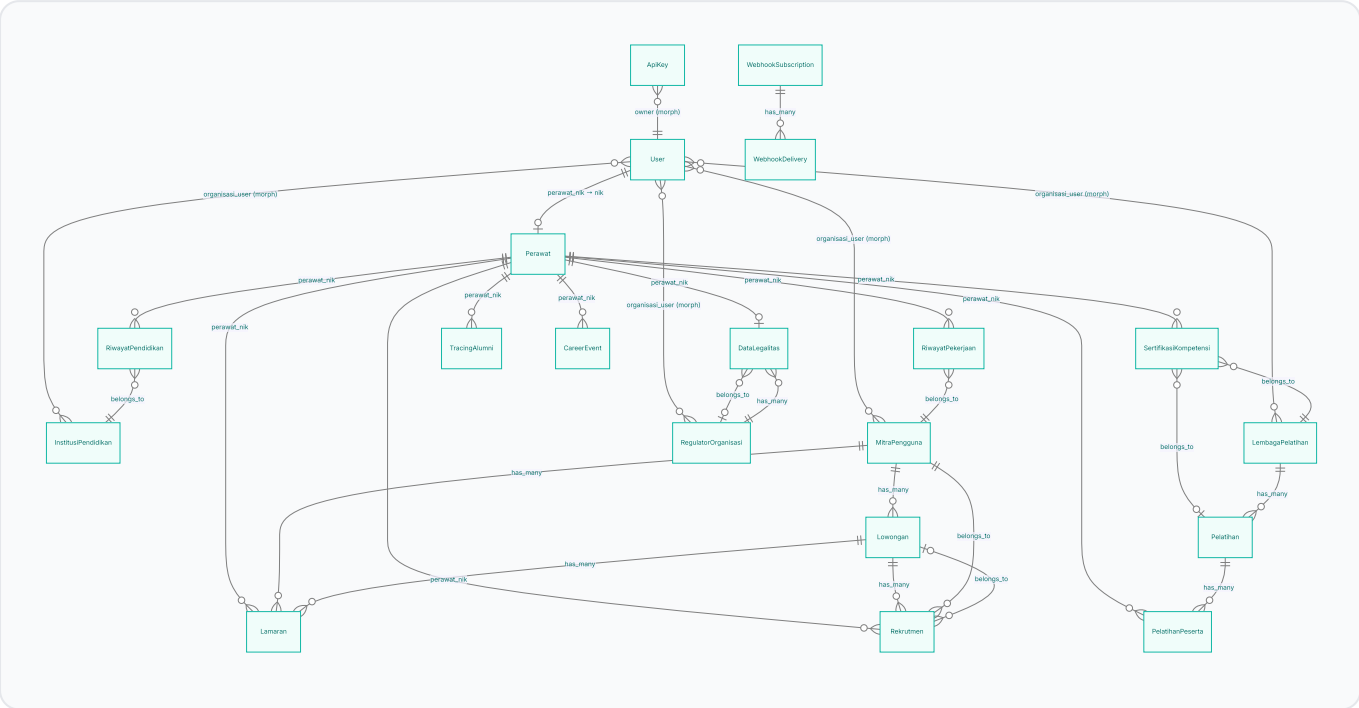
NAMESPACE	JUMLAH
Admin	16 controllers
Api	3 controllers
Auth	Standard Breeze
Kampus	2 controllers
Lembaga	3 controllers
Mitra	7 controllers
Perawat	8 controllers
Regulator	5 controllers

Support Layer

KOMPONEN	JUMLAH
Models	31
Services	22
Policies	7
Notifications	8
Observers	3
Jobs	3
Middleware	4
Migrations	59

2 Entity Relationship Diagram (ERD)

Berikut adalah bagan alir visual (ERD) yang memetakan hubungan antar tabel model pada Platform Keperawatan:

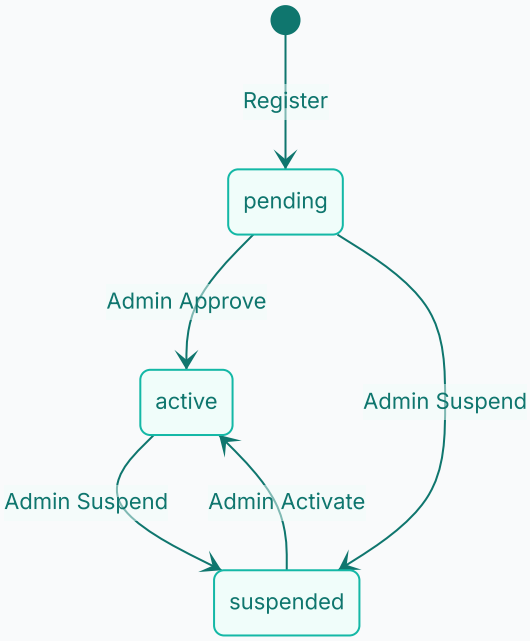


3 Sistem Peran (Role-Based Access Control)

3.1 Definisi 8 Peran

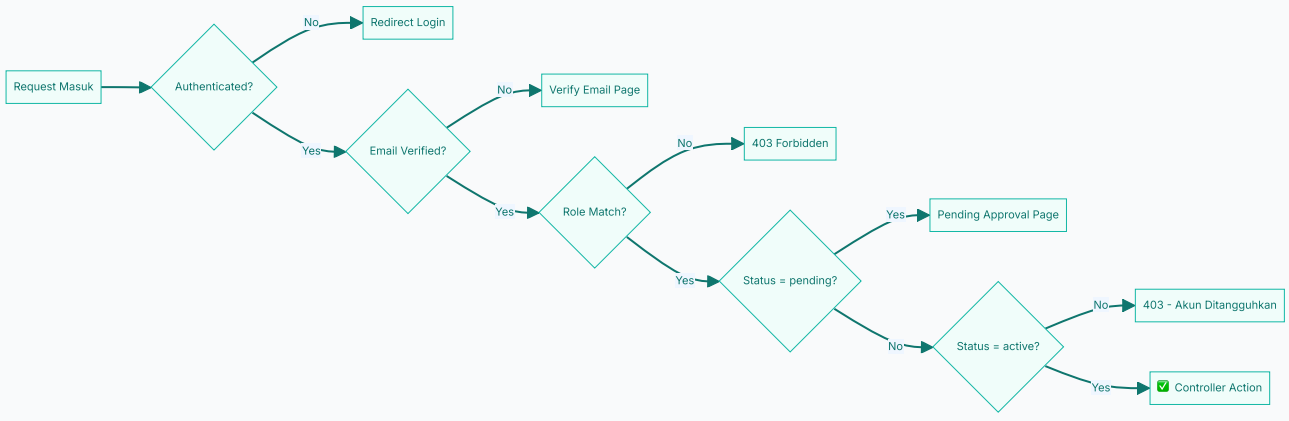
#	ROLE	DESKRIPSI	MIDDLEWARE GUARD	PREFIX URL
1	`perawat`	Tenaga perawat / nurse	role:perawat	/perawat
2	`mitra`	Mitra pengguna / employer	role:mitra,faskes,headhunter	/mitra
3	`faskes`	Fasilitas kesehatan	role:mitra,faskes,headhunter	/mitra
4	`headhunter`	Agen perekrut	role:mitra,faskes,headhunter	/mitra
5	`kampus`	Institusi pendidikan keperawatan	role:kampus	/kampus
6	`lembaga`	Lembaga pelatihan & sertifikasi	role:lembaga	/lembaga
7	`regulator`	Regulator (PPNI, Kemenkes)	role:regulator	/regulator
8	`admin`	Super administrator platform	role:admin	/admin

3.2 Status User — State Machine



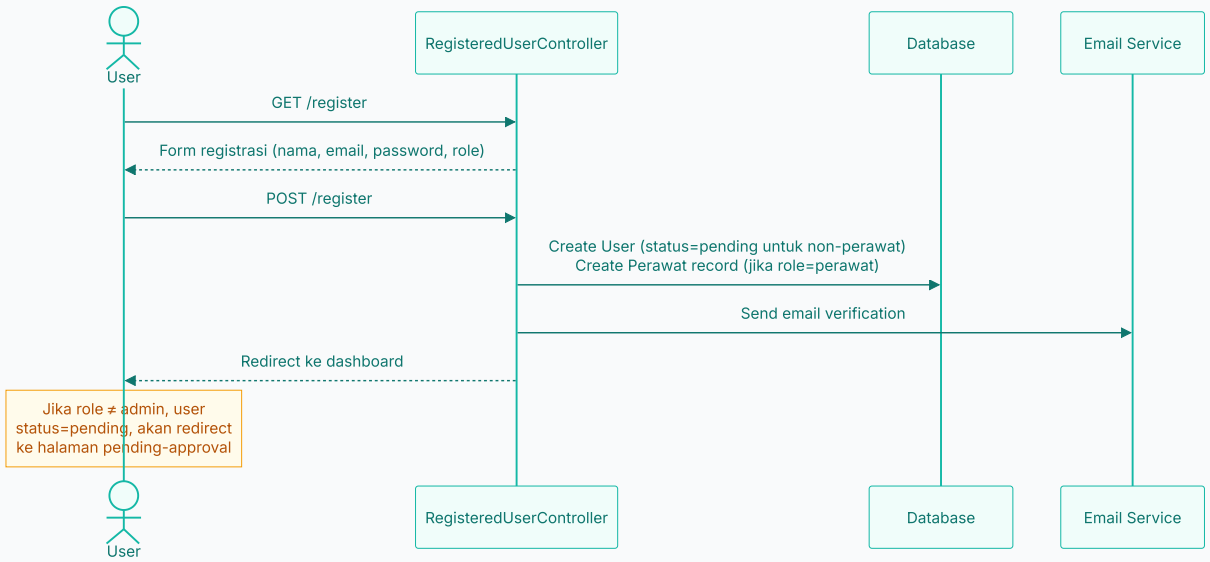
STATUS	AKSES	BEHAVIOR
PENDING	Terbatas	Redirect ke halaman "Menunggu Persetujuan Admin"
ACTIVE	Penuh	Akses penuh sesuai role
SUSPENDED	Diblokir	403 Forbidden — "Akun Anda ditangguhkan"

3.3 Middleware Chain Flow

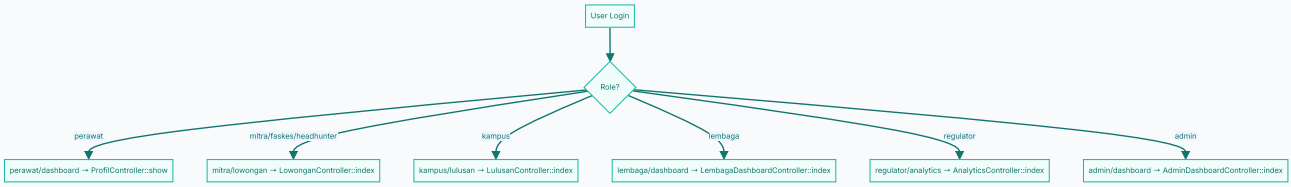


4 Alur Autentikasi & Otorisasi

4.1 Registrasi



4.2 Login → Dashboard Routing



4.3 Auth Routes

GET	/register	Form registrasi
POST	/register	Submit registrasi (throttle: 10/min)
GET	/login	Form login
POST	/login	Submit login (throttle: 20/min)
GET	/forgot-password	Form forgot password
POST	/forgot-password	Send reset link (throttle: 5/min)
GET	/reset-password/{token}	Form reset password
POST	/reset-password	Submit new password
GET	/verify-email	Email verification prompt
POST	/logout	Logout

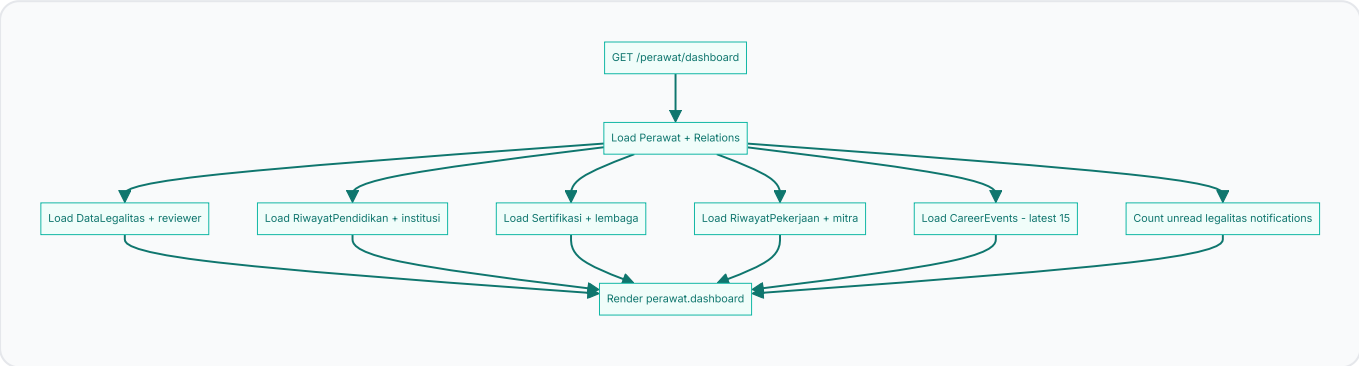
5 Flow per Role — Detail Lengkap

5.1 🩺 PERAWAT (Nurse)

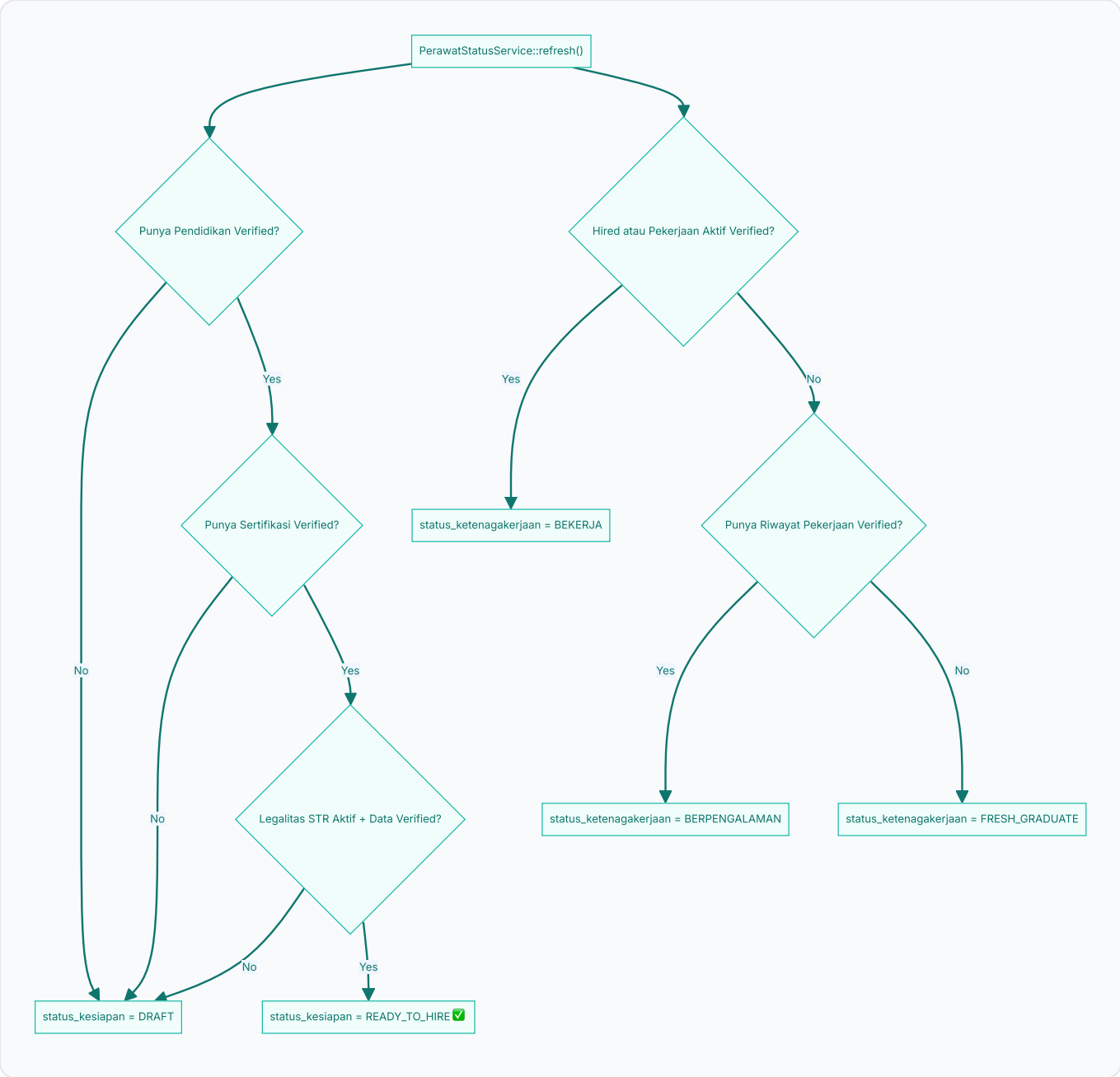
5.1.1 Dashboard Relations & Status Kesiapan

> **Penting**
>
> Perawat harus memiliki **3 hal terverifikasi** untuk menjadi "Ready to Hire": (1) Riwayat Pendidikan verified (dari Kampus), (2) Sertifikasi Kompetensi verified (dari Lembaga), (3) Data Legalitas STR aktif & data verified (dari Regulator).

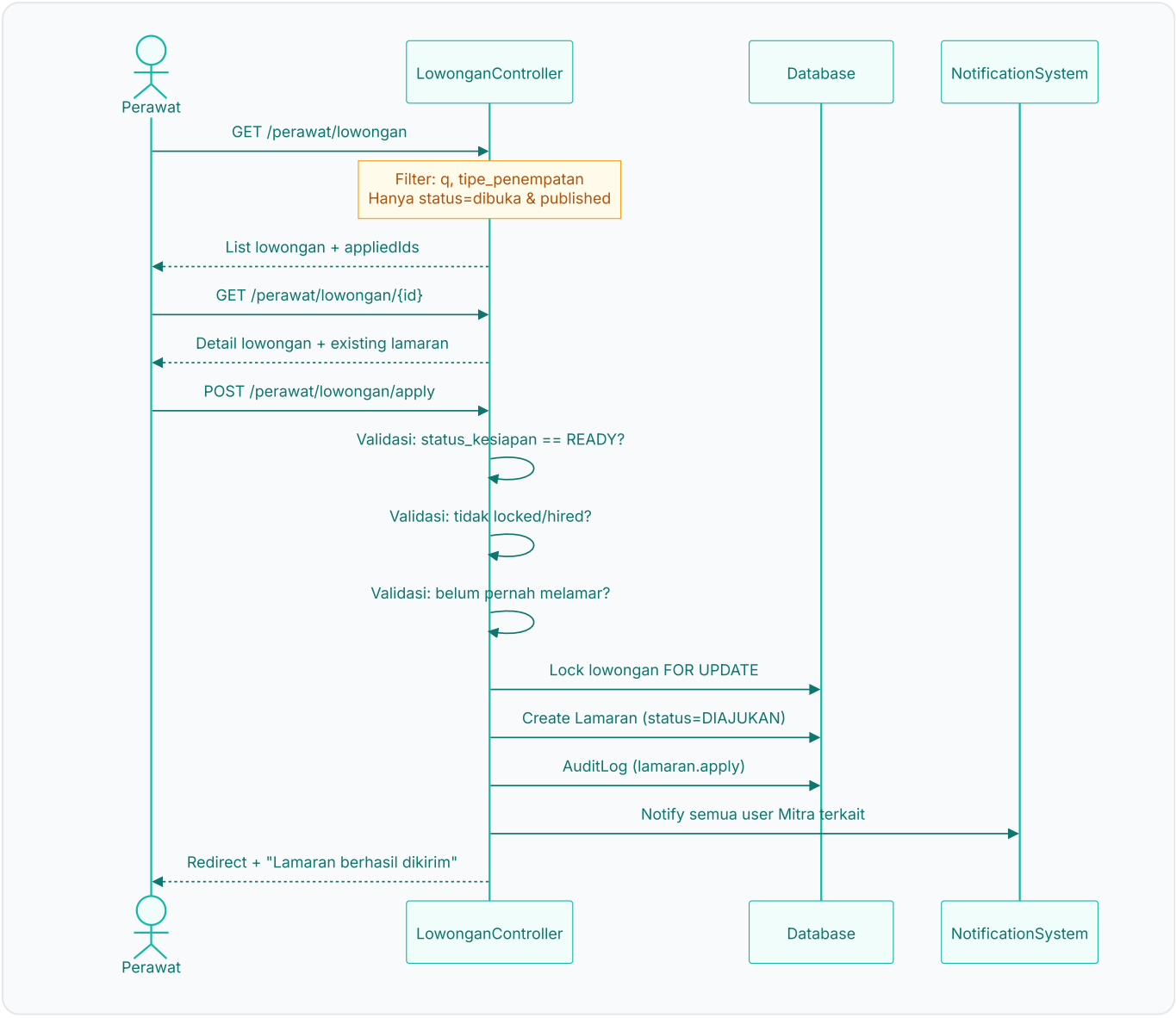
Dashboard Relation Loading



Kesiapan & Ketenagakerjaan Status Determination


PerawatStatusService::refresh()
Punya Pendidikan Verified?
Hired atau Pekerjaan Aktif Verified?
Yes
Yes
No
Punya Sertifikasi Verified?
status_ketenagakerjaan = BEKERJA
Punya Riwayat Pekerjaan Verified?
No
Yes
Yes
No
No
Legalitas STR Aktif + Data Verified?
Yes
No
status_ketenagakerjaan = BERPENGALAMAN
status_ketenagakerjaan = FRESH_GRADUATE
No
Yes
status_kesiapan = DRAFT
status_kesiapan = READY_TO_HIRE ✅

5.1.2 Melamar Lowongan

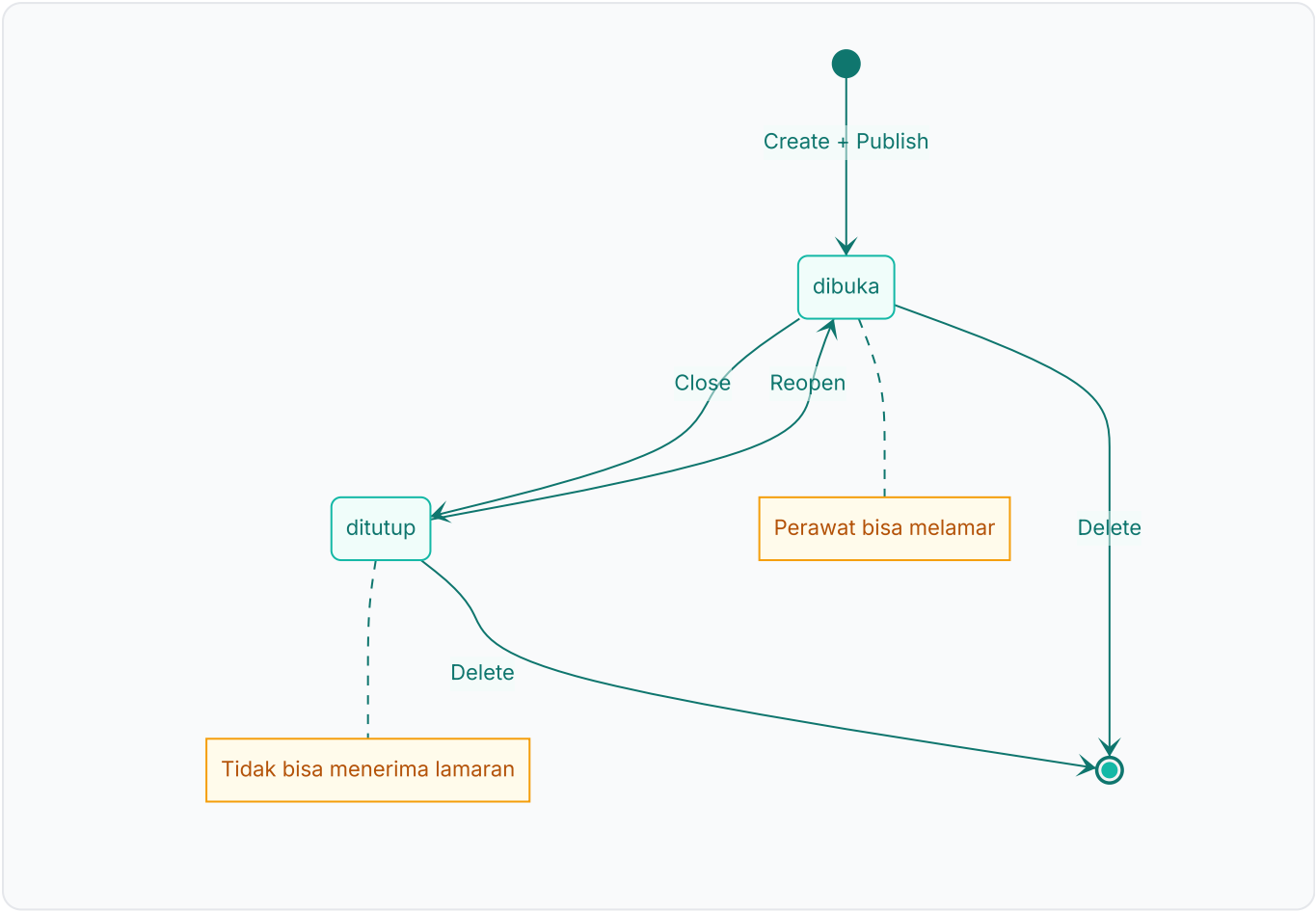


5.1.3 Semua Routes Perawat

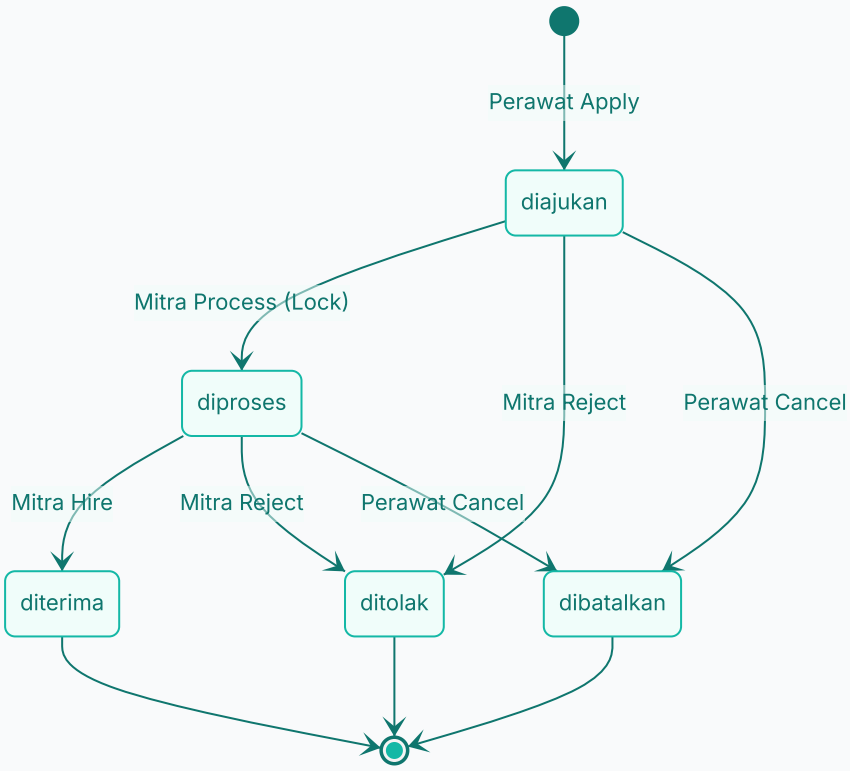
GET	/perawat/dashboard	Dashboard profil + career timeline
GET	/perawat/profil/edit	Form edit data pribadi
PATCH	/perawat/profil	Update profil
PATCH	/perawat/profil/share	Toggle profil publik + generate token
GET	/perawat/cv	Download CV HTML
GET	/perawat/cv-pdf	Download CV PDF
GET	/perawat/lowongan	Browse lowongan (status=dibuka)
GET	/perawat/lowongan/{id}	Detail lowongan
POST	/perawat/lowongan/apply	Melamar lowongan
GET	/perawat/lamaran	List lamaran saya
POST	/perawat/lamaran/{id}/cancel	Batalkan lamaran
GET	/perawat/riwayat-pekerjaan	List riwayat pekerjaan
GET	/perawat/riwayat-pekerjaan/create	Form tambah riwayat (self-report)
POST	/perawat/riwayat-pekerjaan	Store riwayat (verifikasi=pending)
GET	/perawat/riwayat-pekerjaan/{id}/edit	Edit riwayat (jika pending/rejected)
PUT	/perawat/riwayat-pekerjaan/{id}	Update riwayat
DELETE	/perawat/riwayat-pekerjaan/{id}	Hapus riwayat (jika belum verified)
GET	/perawat/str-sync	Form request sync STR
POST	/perawat/str-sync	Submit STR sync request
GET	/perawat/pelatihan	List pelatihan tersedia
GET	/perawat/pelatihan/saya	Pelatihan yang diikuti
POST	/perawat/pelatihan/daftar	Daftar pelatihan
GET	/perawat/notifikasi	List notifikasi
POST	/perawat/notifikasi/mark-read	Tandai semua sudah dibaca
GET	/perawat/legalitas/audit	Audit trail perubahan legalitas

5.2 🏢 MITRA / FASKES / HEADHUNTER (Employer)

5.2.1 Lowongan — State Machine

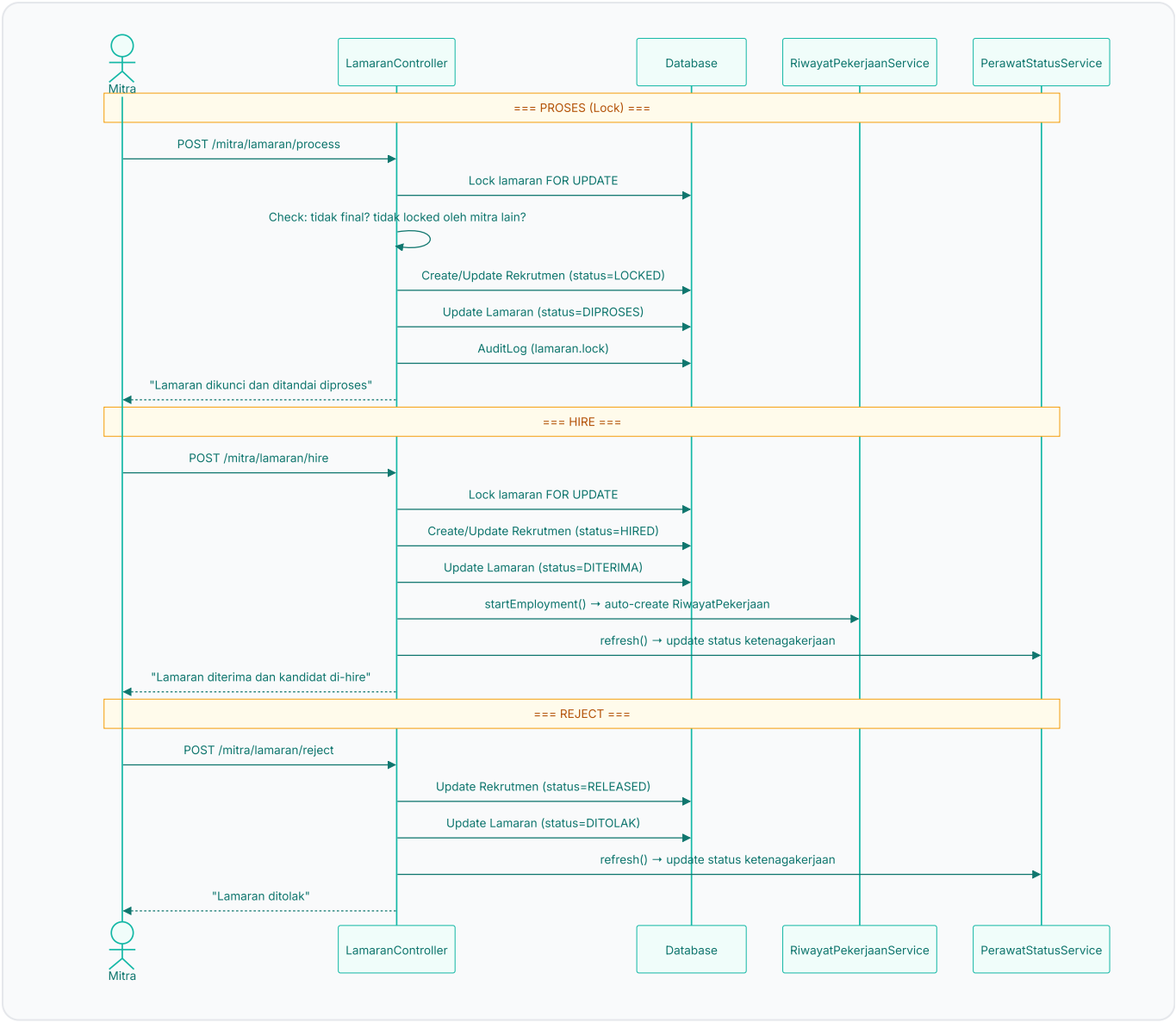


5.2.2 Lamaran — State Machine & Pipeline



> ⚠ **Anti-Poaching Mechanism**
>
> Kandidat yang sedang **LOCKED** atau **HIRED** oleh mitra lain TIDAK bisa diproses. Sistem return HTTP 409 Conflict.

Pipeline Flow: Lock, Hire & Reject


Mitra
LamaranController
Database
RiwayatPekerjaanService
PerawatStatusService
=== PROSES (Lock) ===
POST /mitra/lamaran/process
Lock lamaran FOR UPDATE
Check: tidak final? tidak locked oleh mitra lain?
Create/Update Rekrutmen (status=LOCKED)
Update Lamaran (status=DIPROSES)
AuditLog (lamaran.lock)
"Lamaran dikunci dan ditandai diproses"
=== HIRE ===
POST /mitra/lamaran/hire
Lock lamaran FOR UPDATE
Create/Update Rekrutmen (status=HIRED)
Update Lamaran (status=DITERIMA)
startEmployment() → auto-create RiwayatPekerjaan
refresh() → update status ketenagakerjaan
"Lamaran diterima dan kandidat di-hire"
=== REJECT ===
POST /mitra/lamaran/reject
Update Rekrutmen (status=RELEASED)
Update Lamaran (status=DITOLAK)
refresh() → update status ketenagakerjaan
"Lamaran ditolak"
Mitra
LamaranController
Database
RiwayatPekerjaanService
PerawatStatusService

5.2.3 Pencarian Kandidat — 16 Filter

FILTER	KETERANGAN	TIPE
`nama`	Pencarian nama perawat	LIKE
`jenis_kelamin`	Filter gender	Exact
`status_str`	Status STR (aktif/hold/expired)	Exact
`status_ketenagakerjaan`	fresh_graduate / bekerja / berpengalaman	Exact
`status_ketersediaan`	available / open_to_offer / not_available	Exact
`kota` , `provinsi`	Lokasi perawat	Exact
`jenjang`	Jenjang pendidikan (D3/S1/Ners/S2)	Exact
`program_studi`	Program studi	LIKE
`min_ipk`	IPK minimum	≥ float
`sertifikat`	Nama sertifikat (verified only)	LIKE
`sertifikat_verifikasi`	Exact match sertifikat terverifikasi	Exact
`min_pengalaman`	Minimal tahun pengalaman per record	≥ years
`min_pengalaman_total`	Total akumulasi bulan (overlap merged)	≥ months
`lokasi`	Pencarian alamat / tempat lahir	LIKE

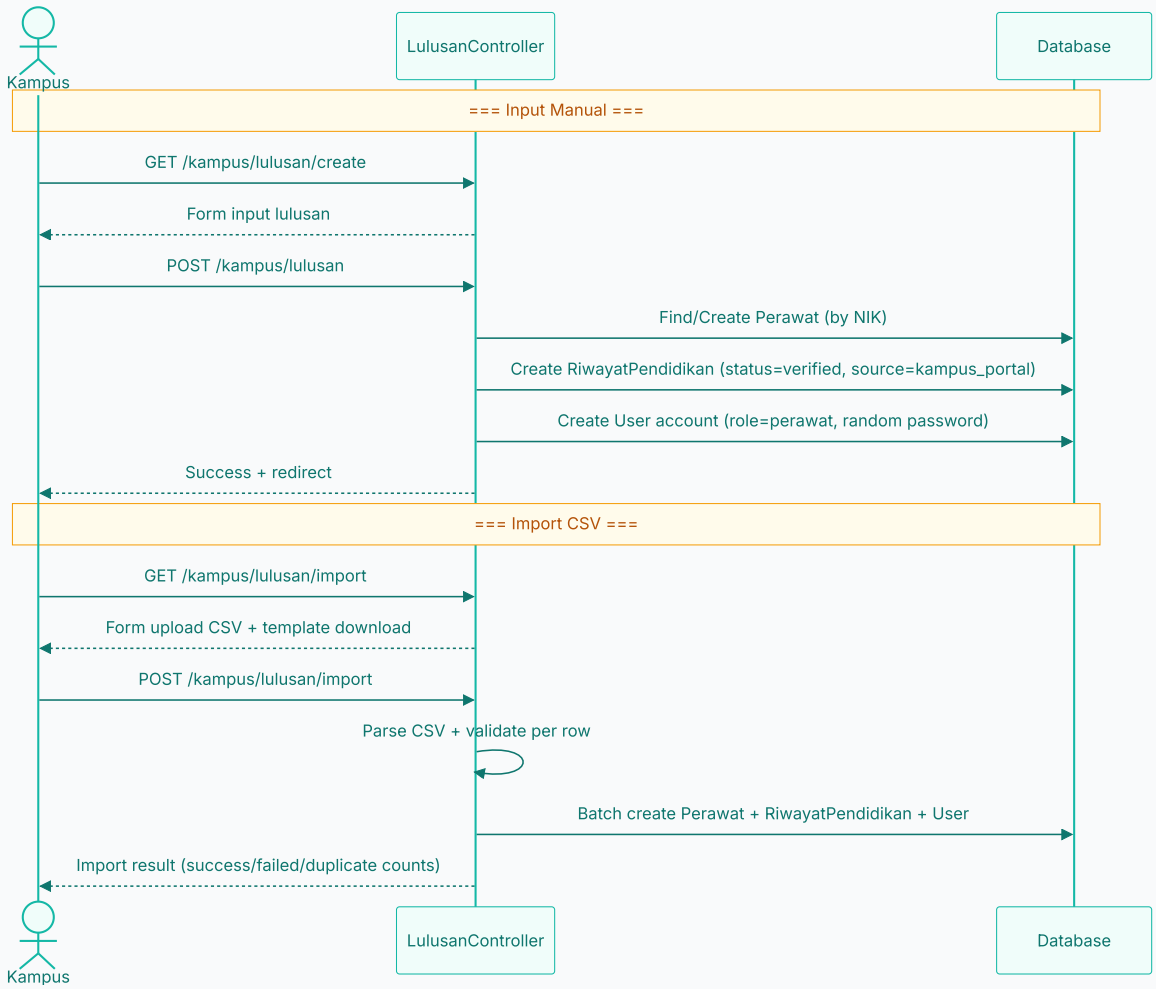
Sorting: `ipk_desc` , `ipk_asc` , `pengalaman_desc` , `pengalaman_asc` , `str_expired` , `str_active` , `nama_asc`

5.2.4 Riwayat Pekerjaan — Verifikasi oleh Mitra

GET	`/mitra/riwayat-pekerjaan/pending`	List pending verifications
GET	`/mitra/riwayat-pekerjaan/aktif`	List active employments
POST	`/mitra/riwayat-pekerjaan/{id}/verify`	Verify riwayat pekerjaan
POST	`/mitra/riwayat-pekerjaan/{id}/reject`	Reject riwayat pekerjaan
POST	`/mitra/riwayat-pekerjaan/{id}/complete`	Mark contract completed

5.3 🎓 KAMPUS (Educational Institution)

5.3.1 Input Lulusan (Manual & CSV Import)



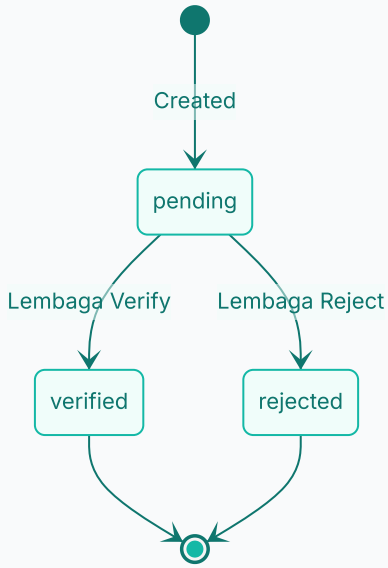
5.3.2 Tracing Alumni

GET	/kampus/tracing-alumni	List alumni + status kerja
GET	/kampus/tracing-alumni/{id}	Detail + form update status
PUT	/kampus/tracing-alumni/{id}	Update status pekerjaan alumni
GET	/kampus/tracing-analytics	Dashboard analytics tracing

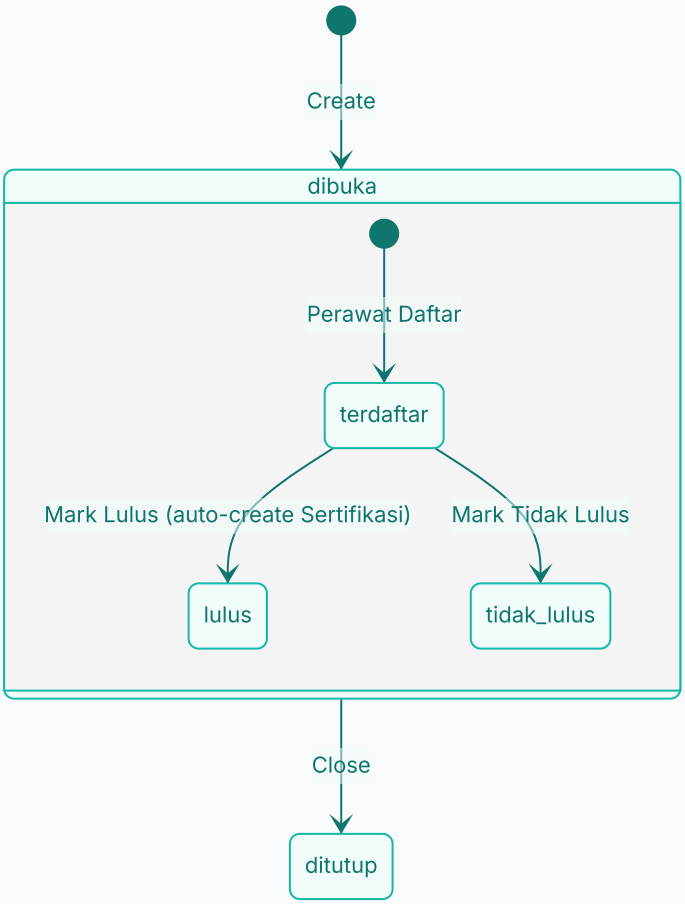
Status Alumni: `belum_bekerja` | `bekerja` | `wirausaha` | `lanjut_studi`

5.4 🏫 LEMBAGA (Training Institution)

5.4.1 Sertifikasi — Status Flow

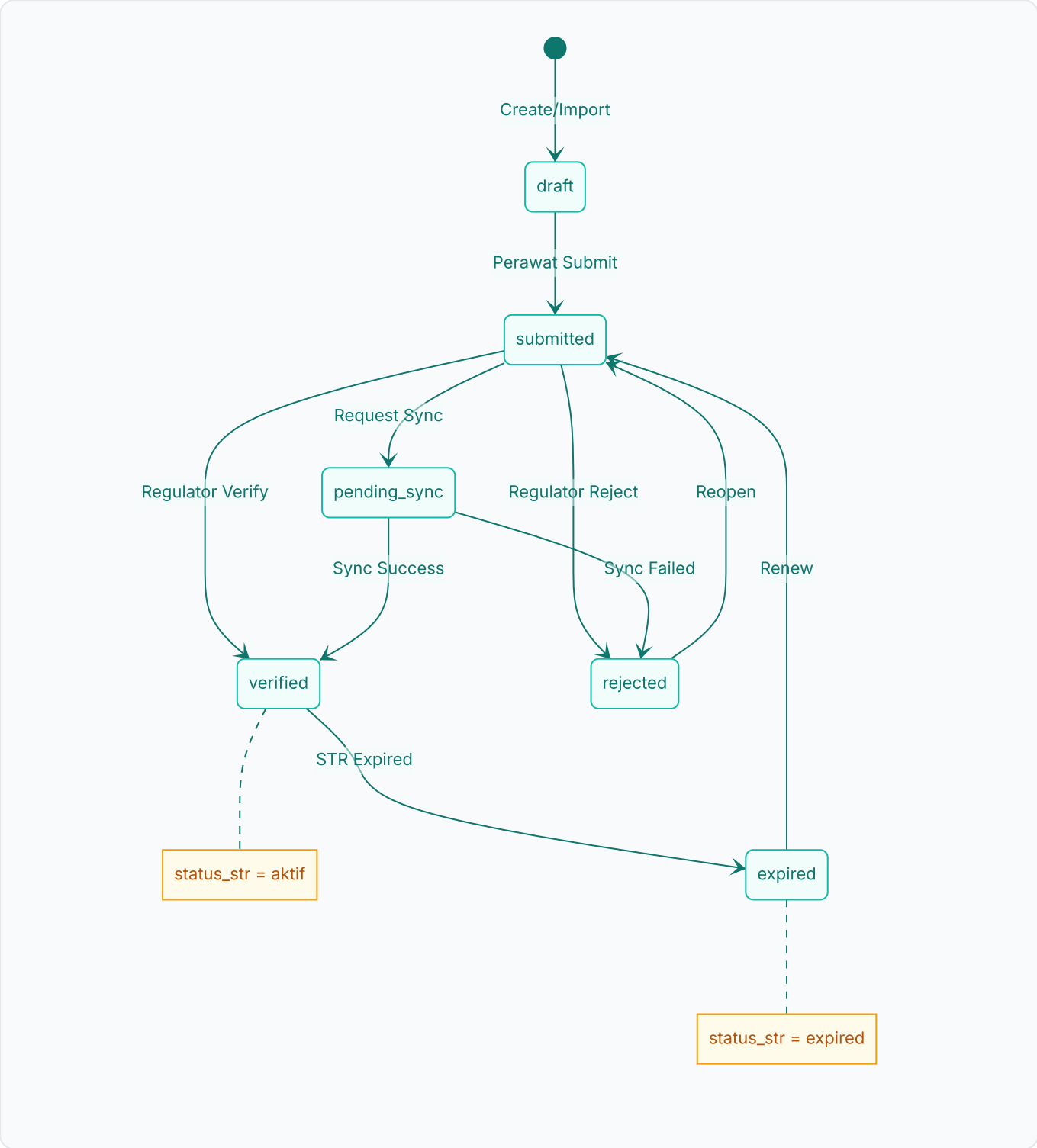


5.4.2 Pelatihan — Auto-create Sertifikasi

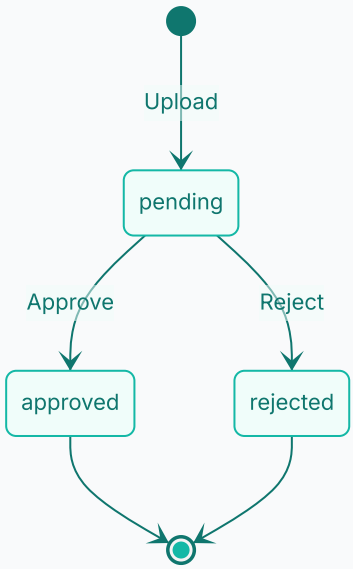


5.5 ⚖️ REGULATOR

5.5.1 Data Legalitas STR — 6 Status State Machine



5.5.2 Dokumen STR — Review Flow


Upload
pending
Approve
Reject
approved
rejected

5.6 👤 ADMIN

5.6.1 Dashboard Metrics

KATEGORI	METRICS
Perawat	Total, Ready, Fresh Graduate, Bekerja, Berpengalaman
STR	Aktif, Hold, Expired, Expiring 30 hari, Tanpa expiry date
Lowongan	Dibuka
Rekrutmen	Locked, Hired
Lamaran	Total, Diajukan, Diterima
Pending Review	Riwayat Pekerjaan, Sertifikasi, Legalitas, Users
Movement (range)	Perawat baru, Legalitas reviewed, Pengalaman verified, Hired, Career Events

Date ranges: 7d, 30d, this_month, custom (max 120 hari). Trend chart per kategori career event.

5.6.2 Admin Routes

GET	`/admin/dashboard`	Dashboard statistik lengkap
GET	`/admin/users`	User management (filter: role, status, search)
POST	`/admin/users/{id}/approve`	Approve user (pending → active)
POST	`/admin/users/{id}/suspend`	Suspend user
POST	`/admin/users/{id}/activate`	Reactivate user
POST	`/admin/users/{id}/reset-password`	Send reset password link
POST	`/admin/users/{id}/impersonate`	Login sebagai user lain
GET	`/admin/api-keys`	Manage API keys
GET	`/admin/audit-logs`	View audit logs
GET	`/admin/data-quality`	Data quality dashboard
GET	`/admin/reports/*.csv`	Export CSV (perawat, lowongan, rekrutmen, lamaran, legalitas, riwayat, audit)

6 Public Routes (Tanpa Auth)

GET	/	Welcome / landing page
GET	/lowongan	Public job listing (semua lowongan dibuka)
GET	/lowongan/{id}	Public job detail
GET	/p/{token}	Public nurse profile (via share token 48 char)
GET	/p/{token}/cv	Download public CV HTML
GET	/p/{token}/cv-pdf	Download public CV PDF
GET	/api/v1/docs	Swagger API documentation page

7 REST API v1

7.1 API Endpoints

METHOD	PATH	SCOPE	DESKRIPSI
POST	`/api/v1/lulusan`	`lulusan:write`	Input data lulusan via API
POST	`/api/v1/sertifikasi`	`sertifikasi:write`	Input sertifikasi via API
POST	`/api/v1/legalitas`	`legalitas:write`	Input/update data legalitas via API
GET	`/api/v1/openapi.json`	—	OpenAPI specification (public)

7.2 API Middleware Stack



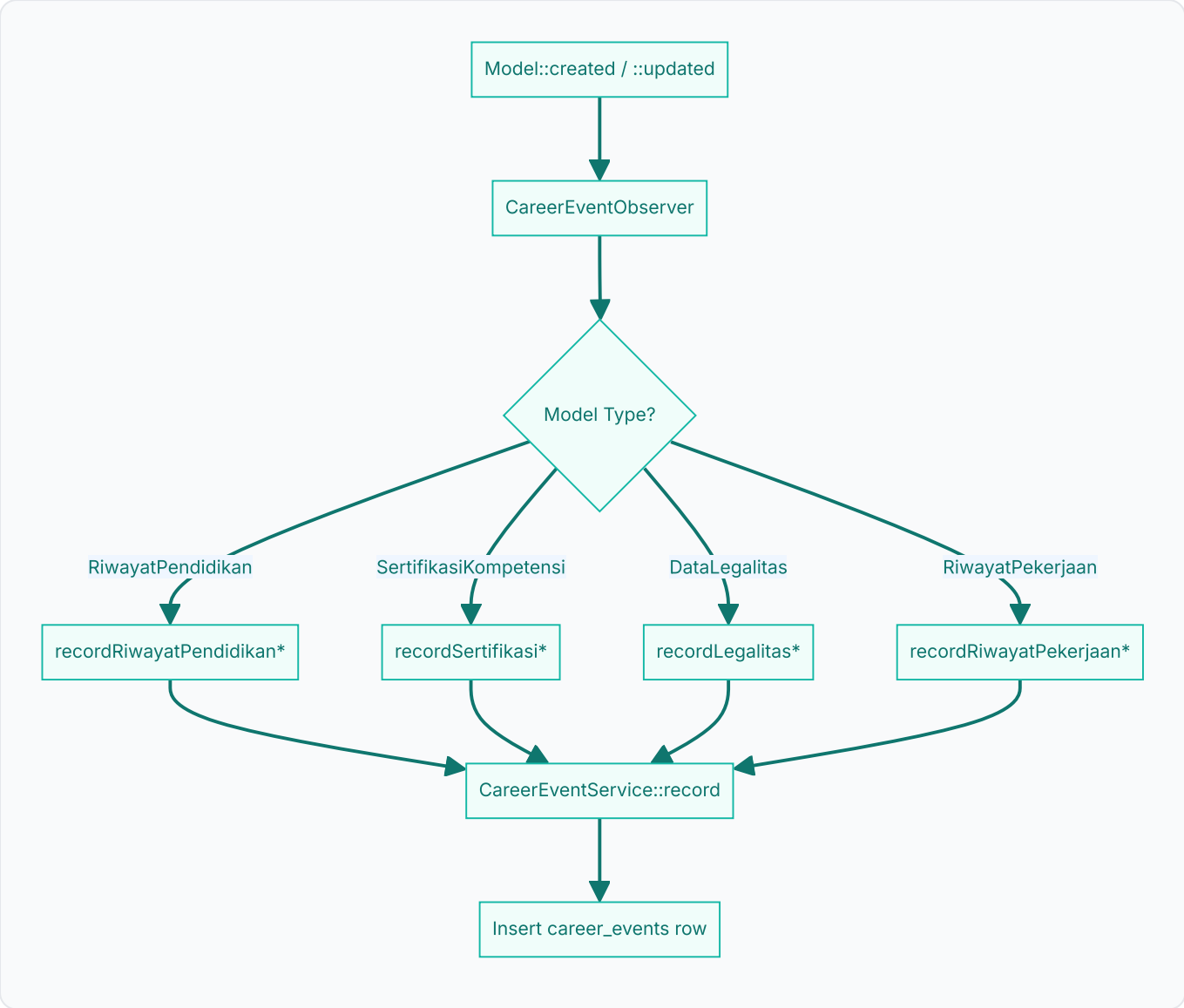
- **ApiKeyAuth** — Validates `X-Api-Key` header + optional HMAC signature via `X-Signature`
- **ApiIdempotency** — Prevents duplicate requests using `Idempotency-Key` header
- **ApiRequestLogger** — Logs every API request for auditing

8 Career Events Timeline

Setiap perubahan signifikan pada data perawat secara otomatis di-record ke tabel `career_events` melalui Eloquent Observer (**CareerEventObserver**).

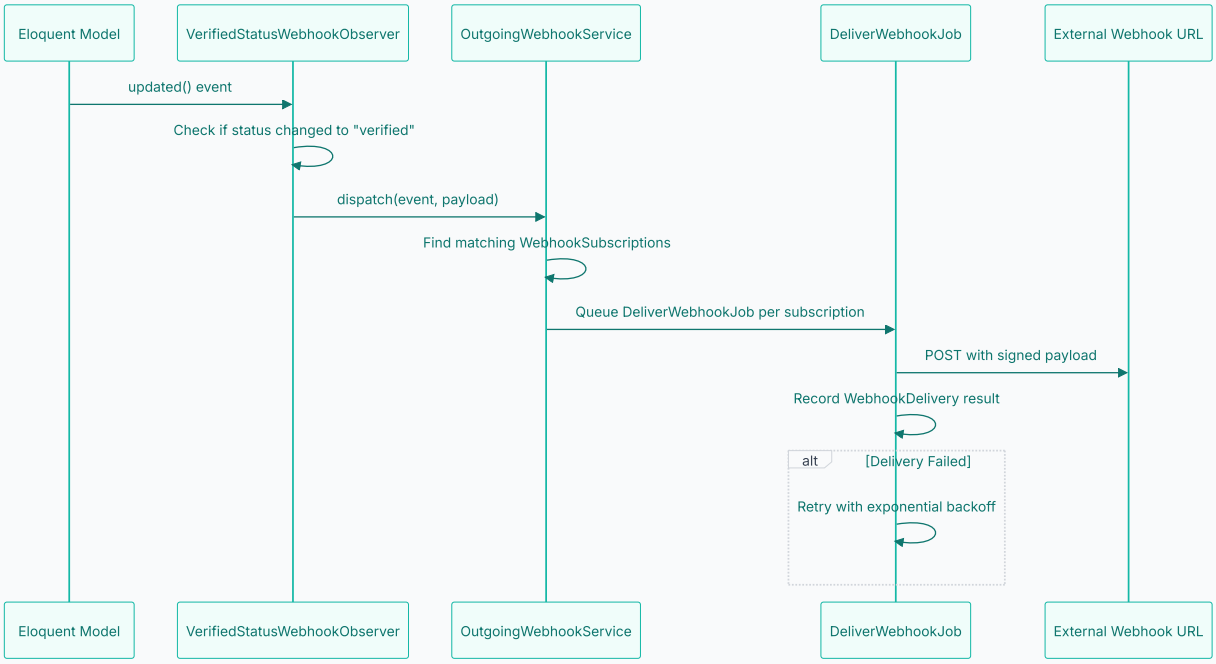
CATEGORY	EVENT KEY	TRIGGER
pendidikan	`lulusan.recorded`	RiwayatPendidikan created
	`lulusan.status_changed`	status_verifikasi berubah
	`lulusan.updated`	Data pendidikan diupdate
sertifikasi	`sertifikasi.recorded`	SertifikasiKompetensi created (pending)
	`sertifikasi.verified`	Status → verified/valid
	`sertifikasi.rejected`	Status → rejected
	`sertifikasi.updated`	Data sertifikasi diupdate
legalitas	`legalitas.draft`	DataLegalitas created (draft)
	`legalitas.submitted`	Data submitted untuk verifikasi
	`legalitas.pending_sync`	Menunggu sinkronisasi
	`legalitas.verified`	STR terverifikasi
	`legalitas.expired`	STR expired
	`legalitas.rejected`	Data ditolak
	`legalitas.updated`	Data legalitas diupdate
pengalaman	`pengalaman.recorded`	RiwayatPekerjaan created
	`pengalaman.submitted`	Diajukan verifikasi (pending)
	`pengalaman.verified`	Pengalaman diverifikasi
	`pengalaman.rejected`	Pengalaman ditolak
	`pengalaman.completed`	Kontrak selesai
	`pengalaman.resigned`	Status resign
kinerja	`kinerja.updated`	Evaluasi kinerja diupdate

8.2 Observer Architecture


Model::created / ::updated
CareerEventObserver
Model Type?
RiwayatPendidikan
SertifikasiKompetensi
DataLegalitas
RiwayatPekerjaan
recordRiwayatPendidikan*
recordSertifikasi*
recordLegalitas*
recordRiwayatPekerjaan*
CareerEventService::record
Insert career_events row

9 Webhook System

9.1 Webhook Delivery Flow



9.2 Webhook Events

EVENT	TRIGGER	PAYLOAD
`legalitas.data.verified`	DataLegalitas `data_status` → `verified`	Perawat NIK, nomor STR, tanggal verifikasi
`sertifikasi.verified`	SertifikasiKompetensi `status_verifikasi` → `verified`	Perawat NIK, nama sertifikat, nomor
`riwayat_pekerjaan.verified`	RiwayatPekerjaan `verifikasi_status` → `verified`	Perawat NIK, posisi, mitra

10 Notification System

NOTIFICATION CLASS	TRIGGER	RECIPIENT
`LowonganBaruNotification`	Lowongan baru dibuat	Perawat matching criteria
`LamaranMasukNotification`	Perawat melamar	User mitra terkait
`LamaranStatusNotification`	Lamaran diproses/ditolak/diterima	User perawat
`LegalitasDataStatusNotification`	Data status legalitas berubah	User perawat
`LegalitasDocumentStatusNotification`	Dokumen STR direview	User perawat
`RiwayatPekerjaanStatusNotification`	Riwayat pekerjaan diverifikasi/ditolak	User perawat
`SertifikasiStatusNotification`	Sertifikasi diverifikasi/ditolak	User perawat
`ExperienceSyncFailedNotification`	Sync pengalaman gagal	User mitra

11 Scheduled Jobs & Artisan Commands

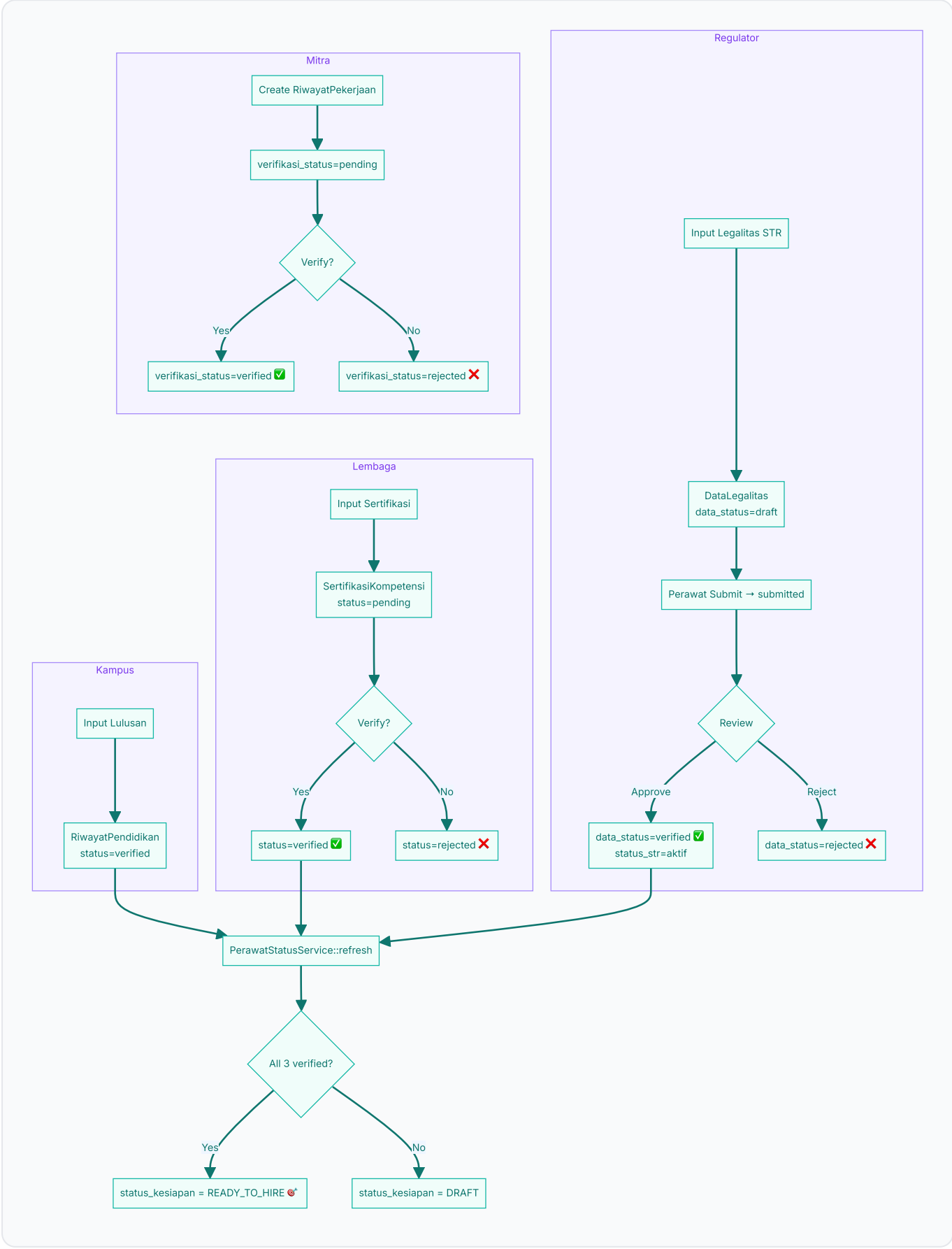
11.1 Scheduled Tasks

SCHEDULE	COMMAND / JOB	DESKRIPSI
Daily 01:00	`legalitas:sync`	Sinkronisasi status legalitas STR dari source eksternal
Daily 02:00	`api:idempotency:purge`	Hapus idempotency key yang expired / stale
Hourly	`SyncExperienceJob`	Sinkronisasi pengalaman kerja dari mitra

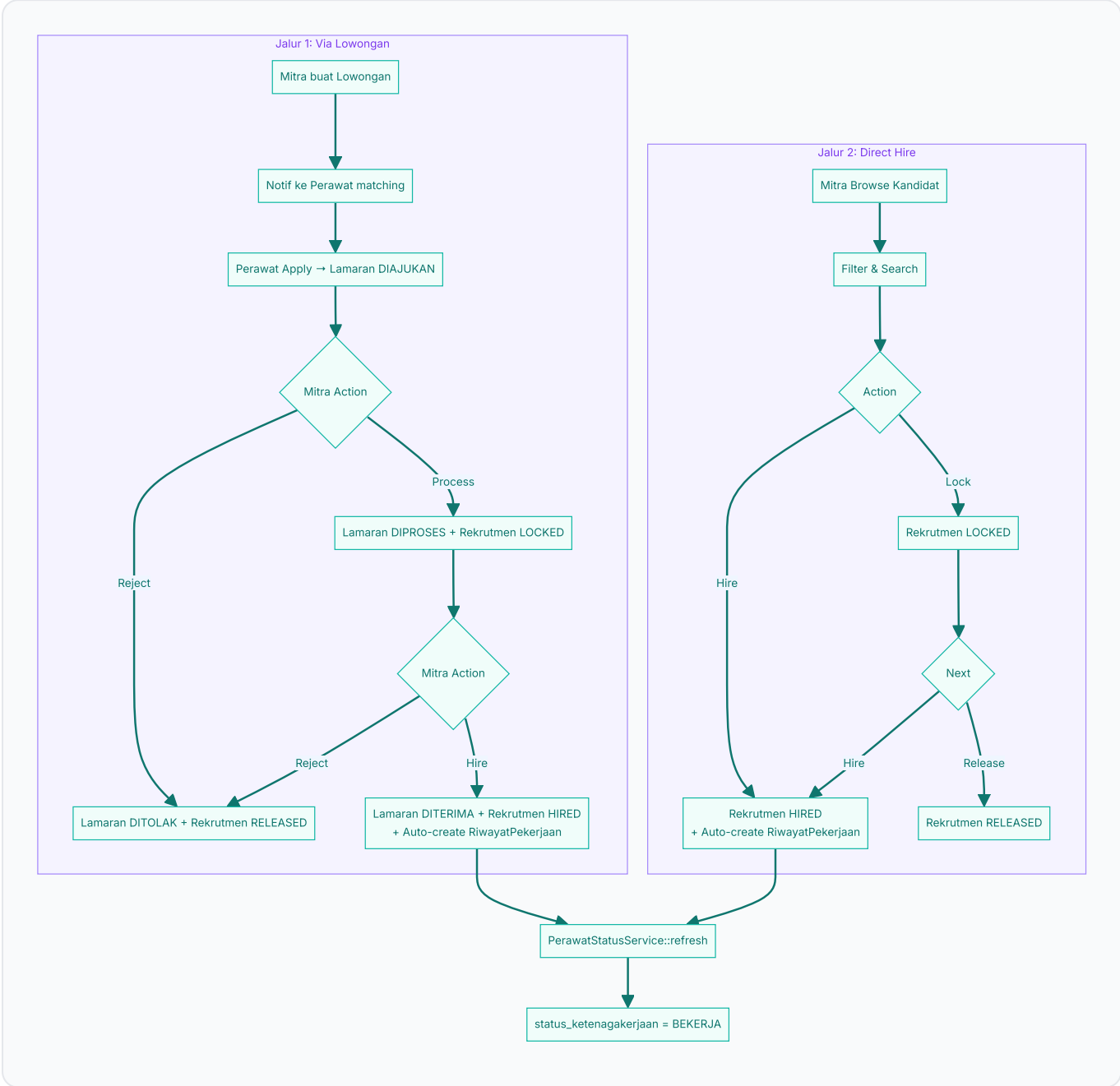
11.2 Artisan Commands

COMMAND	PURPOSE	ENVIRONMENT
`legalitas:sync`	Sync status legalitas STR	All
`experience:sync`	Dispatch job sync pengalaman kerja	All
`api:idempotency:purge`	Purge expired/stale idempotency keys (--dry-run)	All
`demo:refresh`	Reset dan isi ulang demo data (--force)	Non-production
`demo:scenario`	Jalankan timeline demo 1 perawat (--nik=)	Non-production
`preflight:perawat-identity`	Check mismatch perawat_id/nik (--json, --fail-on-warning)	All

12 Data Verification Flow (End-to-End)



13 Recruitment Pipeline (Complete)



14 Identity Model (perawat_nik)

🔑 **Penting: Foreign Key Utama**

Seluruh data portfolio perawat dihubungkan melalui `perawat_nik` (NIK 16 digit) sebagai foreign key, BUKAN `perawat_id`. Ini memungkinkan data tetap terintegrasi meskipun record perawat dibuat secara terpisah oleh berbagai pihak (kampus, lembaga, regulator, mitra).

Tables yang menggunakan perawat_nik sebagai FK

TABLE	RELATIONSHIP	DITULIS OLEH
`users`	User account → Perawat	System (register/kampus import)
`data_legalitas`	STR data → Perawat	Regulator / API
`riwayat_pendidikan`	Education → Perawat	Kampus / API
`sertifikasi_kompetensi`	Certification → Perawat	Lembaga / API
`riwayat_pekerjaan`	Employment → Perawat	Mitra / Perawat
`rekrutmen`	Recruitment → Perawat	Mitra
`lamaran`	Application → Perawat	Perawat
`tracing_alumni`	Alumni tracing → Perawat	Kampus
`pelatihan_peserta`	Training participants → Perawat	System
`career_events`	Timeline → Perawat	System (observer)

15 Source Tracking & Audit

15.1 Provenance Tracking

Setiap record data memiliki field `source_system` dan `source_actor` untuk audit trail:

SOURCE SYSTEM	SOURCE ACTOR	KONTEKS
`kampus_portal`	`kampus`	Data lulusan dari kampus
`lembaga_portal`	`lembaga`	Sertifikasi dari lembaga pelatihan
`regulator_portal`	`regulator`	Data legalitas STR dari regulator
`mitra_portal`	`mitra`	Riwayat pekerjaan dari mitra/employer
`perawat_portal`	`perawat`	Self-reported data dari perawat
`api`	`api_client`	Data via REST API
`integration`	`system`	Auto-sync dari integrasi eksternal

15.2 Audit Log Actions

Lowongan & Lamaran

- `lowongan.create`
- `lowongan.update`
- `lowongan.close` / `lowongan.reopen`
- `lowongan.delete`
- `lamaran.apply`
- `lamaran.lock`
- `lamaran.reject` / `lamaran.hire`

Data & User

- `riwayat_pekerjaan.start`
- `riwayat_pekerjaan.verify` / `.reject`
- `sertifikasi.verify` / `.reject`
- `legalitas.sync` / `.review`
- `user.approve`
- `user.suspend` / `user.activate`
- `user.reset_password`

16 Ringkasan Komponen

31 MODELS	**43** CONTROLLERS	**22** SERVICES	**7** POLICIES
8 NOTIFICATIONS	**3** OBSERVERS	**3** QUEUED JOBS	**4** MIDDLEWARE
59 MIGRATIONS	**140+** WEB ROUTES	**4** API ENDPOINTS	**8** USER ROLES
21 CAREER EVENT TYPES	**3** WEBHOOK EVENTS	**3** SCHEDULED TASKS	**6** ARTISAN COMMANDS

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